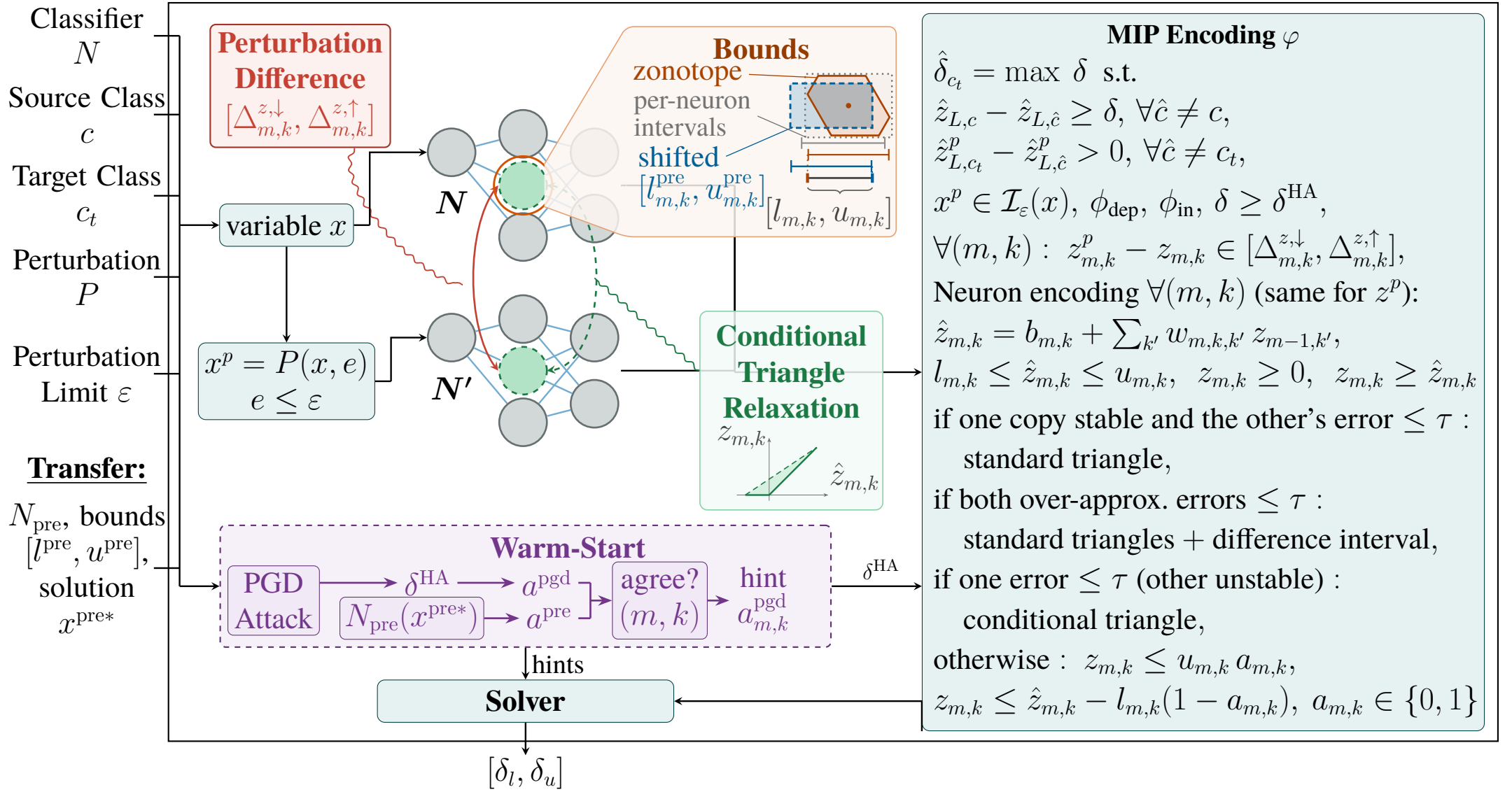

Classifier N
Source Class c
Target Class c_t
Perturbation P
Perturbation Limit ε
Transfer:
N_{pre}, bounds $[l^{\text{pre}}, u^{\text{pre}}]$, solution $x^{\text{pre}*}$
Perturbation Difference
$[\Delta^{z,\downarrow}_{m,k}, \Delta^{z,\uparrow}_{m,k}]$
variable x
$x^p = P(x, e)$
$e \le \varepsilon$
N
N'
Bounds
zonotope
per-neuron intervals
shifted
$[l^{\text{pre}}_{m,k}, u^{\text{pre}}_{m,k}]$
$[l_{m,k}, u_{m,k}]$
Conditional Triangle Relaxation
$z_{m,k}$
$\hat{z}_{m,k}$
Warm-Start
PGD Attack
δ^{HA}
$N_{\text{pre}}(x^{\text{pre}*})$
a^{pgd}
a^{pre}
agree? (m,k)
hint $a^{\text{pgd}}_{m,k}$
δ^{HA}
hints
Solver
$[\delta_l, \delta_u]$
MIP Encoding φ
$\hat{\delta}_{c_t} = \max\ \delta$ s.t.
$\hat{z}_{L,c} - \hat{z}_{L,\hat{c}} \ge \delta,\ \forall \hat{c} \ne c,$
$\hat{z}^p_{L,c_t} - \hat{z}^p_{L,\hat{c}} > 0,\ \forall \hat{c} \ne c_t,$
$x^p \in \mathcal{I}_\varepsilon(x),\ \phi_{\text{dep}},\ \phi_{\text{in}},\ \delta \ge \delta^{\text{HA}},$
$\forall (m,k):\ z^p_{m,k} - z_{m,k} \in [\Delta^{z,\downarrow}_{m,k}, \Delta^{z,\uparrow}_{m,k}],$
Neuron encoding $\forall (m,k)$ (same for z^p):
$\hat{z}_{m,k} = b_{m,k} + \sum_{k'} w_{m,k,k'}\, z_{m-1,k'},$
$l_{m,k} \le \hat{z}_{m,k} \le u_{m,k},\ \ z_{m,k} \ge 0,\ \ z_{m,k} \ge \hat{z}_{m,k}$
if one copy stable and the other's error $\le \tau$:
standard triangle,
if both over-approx. errors $\le \tau$:
standard triangles + difference interval,
if one error $\le \tau$ (other unstable) :
conditional triangle,
otherwise : $z_{m,k} \le u_{m,k}\, a_{m,k},$
$z_{m,k} \le \hat{z}_{m,k} - l_{m,k}(1 - a_{m,k}),\ a_{m,k} \in \{0,1\}$